A-A-53808
19 January 1989

COMMERCIAL ITEM DESCRIPTION

DRILL, TWIST, BONE

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a bone twist drill suitable for all types of orthopedic drilling procedures.

Salient characteristics.

General. The drill shall be 3 inches in length with a 3/32 inch diameter.

Style and design. The drill shall be in accordance with Defense Personnel Support Center (DPSC) Drawing No. 20487. The following dimensions shall apply:

Helix angle: 20 to 24 degrees

Lip relief angle: 12 to 15 degrees

Chisel edge angle: 133 to 137 angle

Material. The drill shall be fabricated from type 316 L corrosion resistant steel.

Hardness. The drill shall have a Rockwell Hardness C scale (HRC) of 35 to 40. The hardness shall be tested in accordance with ASTM Designation E18 or ASTM Designations E92 and E140.

Finish. The finish of the drill shall be mirror, except the operant ends shall have a fine honed finish with sharp cutting edges.

Runout. The total indicated runout shall not exceed .001 per inch when tested as specified herein.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Sterilization. The drills shall be capable of withstanding repeated sterilization via steam, etrylene oxide, or long-term, 10 hour soaking in cold sterilization solution. The drills shall show no evidence of corrosion when autoclaved as specified herein.

Workmanship. The drills shall be free from burrs or other defects which detract from their appearance or impair their serviceability.

Unit. Package (PG). One package containing six drills, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. The drills shall be examined to determine compliance with all requirements contained in this specification.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, inspection level II, AQL 2.5 except for dimensional examination, the inspection level shall be S-2, AQL 2.5. The unit of product shall be one drill.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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Runout. The drill shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to .0002 inch and equipped with a knife-edge foot, a shadowgraph, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the drill, and so positioned that the foot contacts the drill shank nearest the head flutes. The maximum and minimum readings obtained during one complete 360 degrees rotation of the drill shall be measured and recorded to the nearest .0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Autoclaving. The test drills shall be cleaned and placed in an autoclave. The drills shall then be autoclaved at 250 degrees F $\pm$ 5 degrees F and 27 $\pm$ 1 pounds per square inch pressure for a minimum of 5 minutes. After the drills are removed from the autoclave, the entire surface area of the drill shall be examined for evidence of corrosion.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6515-01-246-0182	DRILL, TWIST, BONE, 3/32 INCH Diameter, 3 Inch, 6's

Ordering data (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTEREST:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MP

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITY:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4402

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53308	DRILL, TWIST, BONE

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street City State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426

(TO BE TACH THIS FORM IT ALONG THIS LINE.)

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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